

Lecture 11 — The Bayes Selector

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11.1 Overview

Lectures 9–10 used the fluid LP in two ways. Its dual prices approximate resource opportunity costs, while its primal solution predicts aggregate hindsight actions. A static randomized implementation achieves $O(\sqrt{T})$ additive loss, and the two-value example shows that this square-root gap is real for the natural high-only static policy.

The source of the loss was not a wrong mean. Realized future counts fluctuate around their means on the $\sqrt{T}$ scale. A dynamic policy can observe which side of that fluctuation is occurring and repeatedly update its prediction. We now ask:

If we can predict what hindsight will do, how should we act online?

Here is the argument we will build:

offline-action prediction $\longrightarrow$ compensated coupling $\longrightarrow$ Bayes selection
 $\longrightarrow$ summable disagreements $\longrightarrow O(1)$ regret.

The distinctive idea is not another approximation to a dynamic value function. It is a change in the prediction target:

dynamic programming asks : What is the value of each action?

Bayes Selector asks : What action will hindsight wish I had taken?

11.2 From a point forecast to a confidence region

With τ periods remaining, let $Z_j(\tau)$ be the random future count of type j . Under iid arrivals with probability q_j ,

$$\mathbb{E}[Z_j(\tau)] = \tau q_j.$$

Hoeffding's inequality gives

$$\mathbb{P}(|Z_j(\tau) - \tau q_j| > u) \leq 2 \exp\left(-\frac{2u^2}{\tau}\right). \quad (11.1)$$

A union bound over finitely many types therefore gives a simultaneous confidence region of width on the order of $\sqrt{\tau \log \tau}$.

This resembles UCB, but the uncertain objects are different. In Lectures 4–5, confidence intervals surrounded unknown arm means learned from data. Here the arrival model is known; confidence surrounds the still-unrealized future counts.

A useful heuristic is to subtract a confidence buffer from a fluid action count,

$$L_{ja}(\tau) = [x_{ja}^{\text{fl}}(\tau) - \kappa u(\tau)]^+,$$

and regard positive L_{ja} as evidence that action a will remain available to hindsight. Turning this heuristic into a general theorem requires careful LP sensitivity and tie-breaking. The direct Bayes Selector formulation below avoids treating this informal stability step as a proved fact: it works with the probability that an action disagrees with hindsight.

11.3 Scope of the guarantee

The strong regret conclusion proved below is conditional. We assume:

1. a known stochastic model of future arrivals;
2. finitely many, sufficiently frequent arrival types;
3. bounded rewards and a known horizon;
4. LP sensitivity controlled by the fixed resource matrix, as in the verified finite-type packing theorem stated below;
5. exogenous arrivals: the policy does not itself change the future type distribution.

These conditions are substantive. The value of the result is not that constant regret is automatic, but that it identifies a route to very small regret when offline actions can be predicted reliably.

11.4 The offline-action viewpoint

Consider an online allocation state s : remaining time, remaining capacities, and any other information relevant to feasibility. The current request has type j and action set $\mathcal{A}_j$; in network RM this is usually

$$\mathcal{A}_j = \{\text{accept}, \text{reject}\}.$$

For a realized future sample path ω , let

$$\text{OPT}(s, \omega)$$

be the value of an offline optimal allocation beginning from state s . The offline optimum knows all future arrivals in ω .

We will call a current action *satisfying* when hindsight can take that same action and still remain optimal. Agreement with every offline optimum is unnecessary; if several offline optima tie, agreement with any one of them is enough.

Example 11.1 (One unit of capacity). *A low request worth 1 is here now. If the realized future contains a high request worth 2, every offline optimum rejects the low request, so rejecting is satisfying and accepting is not. If no future request arrives, accepting the low request is satisfying. Thus whether an action is satisfying depends on the realized future that the online policy cannot yet see.*

Definition 1 (Satisfying action). *An action $a \in \mathcal{A}_j$ is satisfying in state s and future realization ω if there exists an offline optimal allocation beginning from (s, ω) that takes action a now.*

The definition allows ties. We do not ask whether every offline optimum takes a ; we ask whether some offline optimum can agree with the online decision without sacrificing hindsight value.

11.5 Compensated coupling

Imagine forcing the offline solution to agree with every online action. Whenever this changes the offline optimum's value, charge the online policy exactly the lost hindsight value as compensation. If these charges telescope over time, total regret becomes the sum of the costs of consequential disagreements. This thought experiment is called *compensated coupling*.

Run the online policy and an offline optimum on the same arrival sequence. At each period:

1. reveal the current type;
2. let the online policy choose its action;
3. if that action is satisfying, select an offline optimum that takes the same action;
4. otherwise, pay the offline benchmark enough compensation to force it to follow the online action.

If the actions agree, no compensation is needed. If they disagree in a consequential way, the compensation measures the lost offline value caused by forcing agreement.

Let C_t be the compensation at time t . Telescoping the coupled benchmark values gives

$$\text{Reg}(T) = \mathbb{E} \left[\sum_{t=1}^T C_t \right]. \quad (11.2)$$

For bounded-reward packing problems, there is a constant $c_{\max}$ such that

$$C_t \leq c_{\max} \mathbf{1}\{a_t \text{ is not satisfying}\}.$$

Therefore

$$\text{Reg}(T) \leq c_{\max} \sum_{t=1}^T \mathbb{P}(a_t \text{ is not satisfying}). \quad (11.3)$$

This is the reason to predict offline actions:

$$\text{control costly disagreement with hindsight} \implies \text{control regret}.$$

11.6 Bayes selection

Condition on the current state s and type j . For action $a \in \mathcal{A}_j$, define its conditional disagreement probability

$$q_t(s, j, a) = \mathbb{P}(a \text{ is not satisfying for offline} \mid s, J_t = j). \quad (11.4)$$

The ideal Bayes Selector chooses

$$a_t \in \arg \min_{a \in \mathcal{A}_j} q_t(s, j, a). \quad (11.5)$$

The word “Bayes” refers to averaging over the stochastic future conditional on current information. Unlike UCB, the selector is not learning an unknown reward distribution. It is predicting an offline action under a probabilistic model of future arrivals.

11.7 The two-value example revisited

Recall the one-resource setting from Lectures 9–10. There are τ future periods, remaining capacity b , high requests worth 2, and low requests worth 1. Let

$$H_\tau \sim \text{Binomial}(\tau, 1/2)$$

count future high requests.

For a current low request:

- if we accept, offline disagrees when $H_\tau \geq b$, because the low request displaces a future high;
- if we reject, offline disagrees when $H_\tau \leq b - 1$, because there was enough capacity for the low request after all.

Thus

$$q_{\text{accept}}(\tau, b) = \mathbb{P}(H_\tau \geq b), \quad (11.6)$$

$$q_{\text{reject}}(\tau, b) = \mathbb{P}(H_\tau \leq b - 1). \quad (11.7)$$

The ideal Bayes Selector chooses the smaller tail probability.

Notice what it does not compute: it does not recursively calculate $v_\tau(b)$ at every state. It asks a classification question about hindsight:

On which side of the current action will the clairvoyant usually lie?

11.8 A re-solved LP Bayes Selector

Exact disagreement probabilities such as (11.6)–(11.7) may be difficult to compute in a network. Vera and Banerjee give a tractable selector based on a re-solved fluid LP.

With τ periods and capacity vector $b(\tau)$ remaining, solve

$$\begin{aligned} \max_{x \geq 0} \quad & \sum_{j=1}^n r_j x_j \\ \text{s.t.} \quad & Ax \leq b(\tau), \\ & x_j \leq \mathbb{E}[Z_j(\tau)] = \tau q_j \quad \text{for } j = 1, \dots, n. \end{aligned} \quad (11.8)$$

Let $x^\text{fl}(\tau)$ be any selected optimal solution. When type j arrives, the *Fluid Bayes Selector* accepts if it is feasible and

$$x_j^\text{fl}(\tau) \geq \frac{\mathbb{E}[Z_j(\tau)]}{2} = \frac{\tau q_j}{2}, \quad (11.9)$$

otherwise it rejects.

Thus the threshold really is one half, but it applies to the fraction of expected future type- j arrivals assigned to acceptance by the current fluid solution. The rule is not an arbitrary majority vote. In the analysis, it minimizes explicit upper bounds on the two disagreement probabilities: accepting when hindsight wants none of type j , and rejecting when hindsight wants essentially all remaining type- j arrivals.

11.9 The clean-event proof pattern

Let $Z(\tau)$ be the future count vector. The re-solved fluid LP uses $\mathbb{E}[Z(\tau)]$, while the coupled offline LP uses the realization $Z(\tau)$ but the *same* remaining capacity $b(\tau)$. LP sensitivity supplies a constant $\kappa(A)$ such that an optimal solution can be selected with

$$\|x^{\text{off}}(Z(\tau)) - x^{\text{fl}}(\tau)\|_{\infty} \leq \kappa(A) \|Z(\tau) - \mathbb{E}[Z(\tau)]\|_1. \quad (11.10)$$

This is the precise stability statement needed by the verified packing theorem; it is not a generic claim that every optimizer varies smoothly.

For a fixed type j , a costly disagreement under (11.9) requires a deviation large enough to move the offline solution across the half-demand threshold. Multinomial concentration therefore gives an exponentially small disagreement probability away from a bounded endgame. More generally, a bound of the form

$$q_{\tau} \leq \frac{C_0}{\tau^2}$$

is enough, because

$$\sum_{\tau=1}^T q_{\tau} \leq C_0 \sum_{\tau=1}^{\infty} \frac{1}{\tau^2} < \infty. \quad (11.11)$$

Compensated coupling then converts the finite total disagreement probability into finite expected regret.

11.10 Verified constant-regret guarantees

The first result gives a clean explicit constant for the one-resource multi-secretary problem. Suppose there are n finite types, type j arrives iid with probability $q_j > 0$, rewards are nonnegative and at most $r_{\max}$, and every accepted request uses one unit of capacity. Let $q_{\min} = \min_j q_j$.

Theorem 11.2 (Fluid Bayes Selector: multi-secretary). *For multinomial arrivals, the Fluid Bayes Selector in (11.9) satisfies*

$$\text{Reg}(T) \leq 2r_{\max} \sum_{j>1} \frac{1}{q_j} \leq \frac{2(n-1)r_{\max}}{q_{\min}}. \quad (11.12)$$

The bound is independent of the horizon T and the initial capacity.

The index $j > 1$ uses the paper's ordering of the reward types; one type is excluded from the sum. The cruder second bound is often easier to remember.

For a d -resource online packing problem with a fixed resource matrix A , the same paper proves the network statement

$$\text{Reg}(T) \leq 103 d r_{\max} \kappa(A)^2 \sum_{j=1}^n \frac{1}{q_j} \quad (11.13)$$

under multinomial arrivals. Here $\kappa(A)$ is the LP sensitivity constant in (11.10). Again, the right-hand side depends on the fixed instance but not on T or the initial resource vector.

Remark 11.3 (Continuous values: a separate later result*). *The theorems above assume finitely many types. The following continuous-value result is not a theorem from the Bayes Selector paper. In later work, Robert Bray shows that constant regret generally fails with continuous values: even the uniform $[0, 1]$ multisecretary problem has regret of order $\log T$. For that setting, Bray proves the explicit upper bound*

$$\text{Reg}(T) \leq \frac{\log(T+1) + 7}{8}$$

and a matching logarithmic lower bound (up to constants) when capacity is a fixed fraction of the horizon. More general continuous-type statements need additional regularity beyond merely having a bounded density. Thus the finite-type conclusion above is order 1 in T , while the natural continuous analogue is typically logarithmic rather than constant.

The result is strong because the benchmark is the offline clairvoyant. It is conditional because finite types, positive type probabilities, bounded rewards, exogenous arrivals, and controlled LP sensitivity are doing real work.

Returning to the three scales from Lectures 9–10, a fluid policy can capture order- T value while losing order- $\sqrt{T}$. Under the stated conditions, Bayes selection uses concentration and stable offline predictions to target the order-1 scale:

Bayes selection uses scale to go beyond first-order optimality.

11.11 UCB and Bayes Selector

The lectures now share a common proof architecture:

	UCB	Bayes Selector
uncertain object	arm mean μ_i	future arrivals Z
confidence object	interval around μ_i	region around Z
decision rule	optimistic arm	offline-predicted action
key implication	$\Delta_i \leq 2r_i$	large deviation or bounded endgame
regret control	few bad pulls	few costly disagreements

In UCB, a suboptimal arm can be selected only while uncertainty is large. In Bayes Selector, an online action can disagree with hindsight only when confidence fails, the optimization is unstable, or the system is in a bounded endgame.

The final equation to remember is

$$\boxed{\text{probabilistic clean event} \implies \text{deterministic control of bad actions} \implies \text{regret bound.}}$$

11.12 What can break the conclusion?

Constant regret should not be expected when important types become too rare, the horizon is unknown and cannot be handled robustly, the offline solution is unstable, rewards are unbounded, or the policy changes the future arrival distribution.

The final failure is the bridge to Lecture 12. In revenue management with substitution, availability changes customer choices. Those sales may then be used to re-estimate demand. The arrival model is no longer cleanly exogenous, and confidence around the wrong model cannot repair the problem.

11.13 Takeaways

1. Compensated coupling converts regret into the cost of disagreements with a dynamic offline benchmark.
2. Bayes Selector chooses actions with small conditional disagreement probability.
3. Re-solving updates the fluid prediction using remaining time and capacity.
4. The Fluid Bayes Selector accepts a feasible type j when its re-solved fluid count is at least half of expected remaining type- j demand.
5. Summable disagreement probabilities yield instance-dependent $O(1)$ regret, going beyond first-order fluid optimality.
6. For multinomial multi-secretary, the verified bound is at most $2(n-1)r_{\max}/q_{\min}$; a verified network bound depends on d , A , rewards, and the type probabilities, but not on T or initial capacity.
7. The same clean-event proof architecture appeared in UCB, but the uncertain object and chosen action are different.

Recommended reading

Core conceptual reading.

Vera and Banerjee, *The Bayesian Prophet: A Low-Regret Framework for Online Decision Making*, Sections 3–4, develops satisfying actions, compensated coupling, disagreement probabilities, and Bayes selection. Focus first on the regret-as-compensation identity and the change in prediction target.

Further perspective.

Banerjee and Freund, *Good Prophets Know When the End Is Near*, develops general conditions under which model-predictive control achieves uniform additive loss and clarifies the roles of horizon information and optimization geometry.

Separate continuous-values result*.

Robert L. Bray, *Logarithmic Regret in Multisecretary and Online Linear Programs with Continuous Valuations*. This later paper is the source of the $\Theta(\log T)$ continuous-valuation result above; it is not a Bayes Selector theorem.

Optional / deeper.

These are research-level papers. The compensated-coupling identity, disagreement view-point, and re-solve-and-threshold rule are core; the full technical conditions and proofs of uniform $O(1)$ bounds are optional. Bray's continuous-valuation analysis is also optional.